

ICU & Critical Care

Practice Examination

45 MCQs | 5 Short Questions | 2 Hours | 70 Marks

SECTION A — Multiple Choice Questions (MCQ)

Choose the single best answer for each question. Correct answers are highlighted.

Part 1: Vital Signs & Monitoring

Q1. What is the target oxygen saturation (SpO₂) for a COPD patient on supplemental oxygen?

- A. 94–98%
- B. 88–92% ✓ ANSWER**
- C. 98–100%
- D. 85–88%

Explanation: COPD patients rely on hypoxic drive. Targeting 88–92% prevents suppression of respiratory drive while avoiding hypoxia.

Q2. What is the minimum acceptable Mean Arterial Pressure (MAP) in a critically ill patient?

- A. 55 mmHg
- B. 60 mmHg
- C. >65 mmHg ✓ ANSWER**
- D. >80 mmHg

Explanation: A MAP >65 mmHg is the standard target in sepsis and critical illness to ensure adequate organ perfusion.

Q3. A 70 kg patient is admitted to the ICU. What is the expected normal urine output per hour?

- A. 10–20 ml/hr
- B. 35–70 ml/hr ✓ ANSWER**
- C. 100–150 ml/hr
- D. 5–10 ml/hr

Explanation: Normal urine output is 0.5–1 ml/kg/hr. For 70 kg: $0.5 \times 70 = 35$ ml/hr minimum, $1 \times 70 = 70$ ml/hr maximum.

Q4. A patient opens eyes to voice (3), makes incomprehensible sounds (2), and withdraws from pain (4). What is the GCS score?

A. 7

B. 8

C. 9 ✓ ANSWER

D. 10

Explanation: GCS = Eye (3) + Verbal (2) + Motor (4) = 9. A score <8 generally indicates need for airway protection/mechanical ventilation.

Q5. At what GCS score is endotracheal intubation and mechanical ventilation typically indicated?

A. <10

B. <8 ✓ ANSWER

C. <12

D. <6

Explanation: GCS <8 indicates inability to protect the airway. Intubation is required to prevent aspiration and ensure adequate ventilation.

Part 2: Medical Emergencies

Q6. A patient with DKA is started on insulin. What is the correct initial insulin infusion rate?

A. 0.05 unit/kg/hr

B. 0.1 unit/kg/hr ✓ ANSWER

C. 0.2 unit/kg/hr

D. 1 unit/kg/hr

Explanation: The standard initial insulin infusion in DKA is 0.1 unit/kg/hr as a continuous IV infusion.

Q7. What is the target rate of blood glucose reduction per hour during DKA management?

A. 10–20 mg/dL/hr

B. 50–80 mg/dL/hr ✓ ANSWER

C. 100–150 mg/dL/hr

D. >200 mg/dL/hr

Explanation: Glucose should fall by 50–80 mg/dL/hr. Too rapid a drop risks cerebral edema; too slow suggests inadequate treatment.

Q8. Which of the following is the MOST DANGEROUS complication of DKA treatment?

A. Hypernatremia

B. Cerebral Edema ✓ ANSWER

C. Pulmonary Embolism

D. Hyperkalemia

Explanation: Cerebral edema is the most feared and potentially fatal complication, especially in children with DKA. It results from overly rapid fluid/glucose correction.

Q9. A patient with septic shock has warm extremities, bounding pulse, and low BP. What type of shock is this?

A. Hypovolemic

B. Cardiogenic

C. Obstructive

D. Distributive ✓ ANSWER

Explanation: Septic shock is distributive — massive vasodilation causes low SVR, warm extremities, and hypotension despite normal or increased cardiac output.

Q10. A patient suddenly develops hypotension, absent breath sounds on the left, tracheal deviation to the right, and jugular venous distension. What is the FIRST action?

A. Order a chest X-ray

B. Give IV fluids immediately

C. Needle decompression of the chest ✓ ANSWER

D. Start norepinephrine

Explanation: This is tension pneumothorax (obstructive shock). Immediate needle decompression (2nd intercostal space, midclavicular line) is life-saving — do not wait for imaging.

Q11. What is the FIRST investigation in a patient presenting with acute stroke?

A. MRI brain with contrast

B. CT brain WITHOUT contrast ✓ ANSWER

C. CT angiography

D. Lumbar puncture

Explanation: Non-contrast CT brain is the first-line investigation to differentiate ischemic from hemorrhagic stroke — this distinction determines treatment.

Q12. A patient with acute ischemic stroke presents within 48 hours. BP is 185/100 mmHg. What is the correct management?

A. Aggressively lower BP to <140/90 immediately

B. Maintain BP around 180/105 — do not lower aggressively ✓ ANSWER

C. Give IV labetalol immediately

D. No intervention needed for BP

Explanation: In acute ischemic stroke (first 48 hrs), collateral circulation depends on elevated BP. Aggressive lowering extends the infarct. Target ~180/105 unless thrombolytics are used.

Q13. In hemorrhagic stroke, what is the blood pressure target?

A. <180/120

B. <160/100

C. <140/90 ✓ ANSWER

D. <120/80

Explanation: In hemorrhagic stroke, the target is BP <140/90 to reduce hematoma expansion, unlike ischemic stroke where BP is kept higher.

Q14. Ventricular Fibrillation (V-fib) is identified on the monitor. What is the IMMEDIATE treatment?

A. IV Amiodarone

B. IV Atropine

C. Synchronized DC cardioversion

D. Unsynchronized DC shock (defibrillation) ✓ ANSWER

Explanation: V-fib is a shockable rhythm requiring immediate unsynchronized defibrillation. Synchronized cardioversion is used for organized rhythms (e.g., unstable SVT, VT with pulse).

Q15. DIC (Disseminated Intravascular Coagulation) is characterized by:

A. Only thrombosis

B. Only bleeding

C. Thrombosis AND bleeding simultaneously ✓ ANSWER

D. Neither thrombosis nor bleeding

Explanation: DIC paradoxically causes both clotting (microthrombi) and bleeding (consumption of clotting factors and platelets) at the same time.

Part 3: Critical Care Conditions

Q16. A patient with hepatic encephalopathy has elevated ammonia. Which management approach is correct?

A. High protein diet + IV antibiotics

B. Low protein diet + Lactulose enemas ✓ ANSWER

- C. High protein diet + Lactulose enemas
- D. Nothing by mouth + IV mannitol

Explanation: Lactulose reduces colonic pH, trapping ammonia as NH_4^+ and preventing absorption. Low protein diet reduces substrate for ammonia production.

Q17. In hypothyroidism, which lab finding is expected?

- A. Low TSH, High T4
- B. High TSH, Low T4 ✓ ANSWER**
- C. Low TSH, Low T4
- D. High TSH, High T4

Explanation: In primary hypothyroidism, the thyroid gland is underactive. The pituitary compensates by secreting more TSH. T3/T4 are low while TSH is high.

Q18. A patient has BP of 195/125 mmHg. She is alert, with no headache, chest pain, or neurological signs. This is classified as:

- A. Hypertensive Emergency
- B. Hypertensive Urgency ✓ ANSWER**
- C. Normal hypertension
- D. White coat hypertension

Explanation: Hypertensive Urgency = BP >180/120 WITHOUT acute target organ damage. Management involves gradual BP reduction over 24–48 hours, usually with oral agents.

Q19. Which of the following represents a Hypertensive Emergency?

- A. BP 185/110 in a clinic with no symptoms
- B. BP 190/120 with acute pulmonary edema ✓ ANSWER**
- C. BP 178/115 with mild headache
- D. BP 200/100 with anxiety

Explanation: Hypertensive Emergency = BP >180/120 WITH acute target organ damage (stroke, MI, aortic dissection, acute pulmonary edema, acute kidney injury).

Q20. What is the Cerebral Perfusion Pressure (CPP) formula?

- A. $\text{CPP} = \text{MAP} + \text{ICP}$
- B. $\text{CPP} = \text{MAP} - \text{ICP}$ ✓ ANSWER**
- C. $\text{CPP} = \text{ICP} - \text{MAP}$
- D. $\text{CPP} = \text{MAP} \times \text{ICP}$

Explanation: CPP = MAP – ICP. Target CPP >70 mmHg in head injury. If MAP falls or ICP rises, cerebral perfusion is compromised.

Q21. What is the correct head-of-bed elevation for a mechanically ventilated patient to prevent VAP?

A. 0° (flat)

B. 15–20°

C. >30–45° ✓ ANSWER

D. >60°

Explanation: Elevating the head of bed >30–45° is the cornerstone of VAP bundle — it reduces aspiration of oropharyngeal secretions into the lower respiratory tract.

Q22. A pregnant patient has a BP of 165/108 mmHg with proteinuria and seizures. What is the BP target?

A. <180/120

B. <160/105 ✓ ANSWER

C. <120/80

D. <140/90

Explanation: In eclampsia, the treatment threshold for antihypertensives is BP >160/105 mmHg. The target is to lower BP below this threshold to prevent cerebral complications.

Q23. A patient with suspected acute MI calls from home. While waiting for the ambulance, what should he do?

A. Take metoprolol 50 mg

B. Chew 4 aspirin tablets (300 mg total) ✓ ANSWER

C. Drink water and rest

D. Take his regular antihypertensive

Explanation: Aspirin 300 mg (4 × 75 mg tablets) chewed immediately inhibits platelet aggregation and is the first critical step in ACS management before hospital arrival.

Q24. Which type of shock is characterized by pump failure with low cardiac output?

A. Distributive

B. Obstructive

C. Hypovolemic

D. Cardiogenic ✓ ANSWER

Explanation: Cardiogenic shock = heart failure (pump failure). Low CO, high SVR, cold extremities — opposite to distributive shock.

Q25. The diagnosis of DKA requires which combination of lab findings?

A. Blood glucose >250, pH <7.3, Bicarbonate <15 ✓ ANSWER

B. Blood glucose >180, pH <7.35, Bicarbonate <20

C. Blood glucose >200, pH <7.1, Bicarbonate <10

D. Blood glucose >300, pH <7.4, Bicarbonate <22

Explanation: DKA diagnostic criteria: Blood glucose >250 mg/dL, arterial pH <7.3, and serum bicarbonate <15 mEq/L — plus ketonemia/ketonuria.

Part 4: Additional High-Yield Topics

Q26. In hyperthyroidism (thyrotoxicosis), which lab pattern is expected?

A. High TSH, High T4

B. Low TSH, High T4 ✓ ANSWER

C. High TSH, Low T4

D. Low TSH, Low T4

Explanation: In hyperthyroidism, excess thyroid hormones suppress TSH via negative feedback. Result: Low TSH, High T3/T4.

Q27. What is the most common complication of stroke in the ICU?

A. Deep Vein Thrombosis

B. Aspiration Pneumonia ✓ ANSWER

C. Urinary Tract Infection

D. Pressure sores

Explanation: Aspiration pneumonia is the most common and serious complication in stroke patients due to impaired swallowing and depressed consciousness.

Q28. In ICP management, what head position is recommended?

A. Flat (0°)

B. Trendelenburg (head down)

C. 30–45° elevation, head midline ✓ ANSWER

D. Lateral decubitus

Explanation: Head at 30–45° promotes cerebral venous drainage. Head must be midline (not rotated) to prevent jugular compression and ICP spikes.

Q29. Anaphylactic shock is classified as which type of shock?

- A. Hypovolemic
- B. Cardiogenic
- C. Obstructive

D. Distributive ✓ ANSWER

Explanation: Anaphylaxis causes massive histamine release → systemic vasodilation → distributive shock. Management includes IM epinephrine.

Q30. Which intervention is NOT part of the standard VAP (Ventilator-Associated Pneumonia) prevention bundle?

- A. Oral care every 4–6 hours

B. Daily chest X-ray ✓ ANSWER

- C. DVT prophylaxis
- D. Sedation vacation

Explanation: The 5-component VAP bundle: HOB elevation, oral care, sedation vacation/weaning, DVT prophylaxis, GI stress ulcer prophylaxis. Daily chest X-ray is NOT a bundle component.

Q31. Pulmonary Embolism (PE) causes which type of shock?

- A. Distributive
- B. Hypovolemic
- C. Obstructive ✓ ANSWER
- D. Cardiogenic

Explanation: PE causes mechanical obstruction of the pulmonary vasculature → obstructive shock. Hemodynamic compromise indicates massive PE.

Q32. Normal urine output in an adult is 0.5–1 ml/kg/hr. For a 60 kg patient, this equals:

- A. 10–20 ml/hr
- B. 30–60 ml/hr ✓ ANSWER
- C. 60–100 ml/hr
- D. 100–120 ml/hr

Explanation: $0.5 \times 60 = 30$ ml/hr (minimum). $1 \times 60 = 60$ ml/hr (maximum). Below 30 ml/hr = oliguria in this patient.

Q33. In neurogenic shock (a type of distributive shock), which finding is typical?

- A. Tachycardia and hypertension

B. Bradycardia and hypotension ✓ ANSWER

- C. Tachycardia and hypotension
- D. Bradycardia and normal BP

Explanation: Neurogenic shock (spinal cord injury) causes loss of sympathetic tone → vasodilation (low BP) + unopposed parasympathetic activity → paradoxical bradycardia.

Q34. Which of the following is the BEST indicator of adequate fluid resuscitation in shock?

- A. Heart rate returning to normal
- B. Urine output >0.5 ml/kg/hr
- C. MAP >65 mmHg

D. All of the above ✓ ANSWER

Explanation: Adequacy of resuscitation is assessed using multiple endpoints: normalization of heart rate, MAP >65 mmHg, UO >0.5 ml/kg/hr, and improving mental status/lactate.

Q35. What is the target Cerebral Perfusion Pressure (CPP) in traumatic brain injury?

- A. CPP >50 mmHg
- B. CPP >60 mmHg
- C. CPP >70 mmHg ✓ ANSWER
- D. CPP >80 mmHg

Explanation: CPP >70 mmHg is the target to ensure adequate cerebral blood flow. $CPP = MAP - ICP$, so maintaining high MAP and controlling ICP are both essential.

Q36. A patient in the ICU on a ventilator has fever, new infiltrate on CXR, and purulent tracheal secretions 72 hours after intubation. Diagnosis?

- A. Community-acquired pneumonia
- B. Ventilator-Associated Pneumonia (VAP) ✓ ANSWER
- C. Pulmonary embolism
- D. Tuberculosis

Explanation: VAP is defined as pneumonia occurring >48 hours after intubation. New infiltrate + fever + purulent secretions in a ventilated patient = VAP until proven otherwise.

Q37. In hypovolemic shock, which set of findings is expected?

- A. Low BP, High HR, Low CVP, Cold extremities ✓ ANSWER
- B. Low BP, Low HR, High CVP, Warm extremities
- C. High BP, High HR, Low CVP, Cold extremities
- D. Normal BP, High HR, Normal CVP, Warm extremities

Explanation: Hypovolemic shock: hemorrhage/dehydration → Low intravascular volume → Low BP, reflexive tachycardia, Low CVP (low preload), cold/clammy extremities (vasoconstriction).

Q38. Daily sedation vacation in mechanically ventilated patients is performed to:

A. Prevent DVT formation

B. Assess neurological status and readiness to wean ✓ ANSWER

C. Reduce oxygen consumption

D. Prevent peptic ulcer disease

Explanation: Sedation vacation allows daily assessment of neurological function and readiness for weaning/extubation — a core component of the VAP bundle and liberation from mechanical ventilation.

Q39. Which of the following is an indication to start DVT prophylaxis in an ICU patient?

A. Only if the patient has a known clot

B. Routinely for all mechanically ventilated patients ✓ ANSWER

C. Only if the patient is ambulatory

D. Only after surgery

Explanation: ICU patients are high-risk for DVT due to immobility, critical illness, and central lines. DVT prophylaxis (LMWH or compression devices) is a routine VAP bundle component.

Q40. Hypertensive urgency differs from hypertensive emergency in that urgency:

A. Has higher BP readings

B. Occurs only in young patients

C. Has NO acute target organ damage ✓ ANSWER

D. Requires IV antihypertensives immediately

Explanation: The key distinction: Emergency = BP >180/120 WITH organ damage (brain, heart, kidneys). Urgency = same BP WITHOUT organ damage. Urgency is managed with oral agents over 24–48 hrs.

Q41. Which finding on DKA workup would most indicate the need for potassium supplementation?

A. Serum K⁺ = 5.8 mEq/L

B. Serum K⁺ = 4.2 mEq/L

C. Serum K⁺ = 3.1 mEq/L ✓ ANSWER

D. Serum K⁺ = 5.2 mEq/L

Explanation: Despite total body K⁺ depletion in DKA, serum K⁺ appears falsely normal/high initially. Insulin therapy will drive K⁺ intracellularly, risking life-threatening hypokalemia — supplement if K⁺ <3.5.

Q42. A patient has BP 200/130 with acute vision changes, confusion, and rising creatinine. This is best described as:

- A. Hypertensive urgency
- B. Hypertensive emergency
- C. Malignant hypertension causing end-organ damage

D. Both B and C ✓ ANSWER

Explanation: Malignant hypertension is a severe form of hypertensive emergency with papilledema and multi-organ involvement. It requires immediate IV antihypertensive therapy in ICU.

Q43. Lactulose enemas in hepatic encephalopathy work by:

- A. Decreasing bilirubin levels
- B. Acidifying the colon to trap ammonia as NH_4^+ ✓ ANSWER**
- C. Increasing protein metabolism
- D. Stimulating the liver to regenerate

Explanation: Lactulose lowers colonic pH → converts NH_3 (absorbable) to NH_4^+ (non-absorbable) → trapped and excreted in stool → reduces blood ammonia levels.

Q44. Pulseless Ventricular Tachycardia (pVT) should be treated with:

- A. IV Amiodarone alone
- B. CPR only, no electricity
- C. Immediate unsynchronized DC shock ✓ ANSWER**
- D. Synchronized cardioversion

Explanation: Pulseless VT = shockable rhythm → immediate unsynchronized defibrillation (same as V-fib). Synchronized cardioversion is only for organized rhythms WITH a pulse.

Q45. Which complication is most likely to cause death from DKA in a pediatric patient?

- A. Hypoglycemia
- B. Hypokalemia
- C. Cerebral edema ✓ ANSWER**
- D. Aspiration pneumonia

Explanation: Cerebral edema is the leading cause of DKA-related death in children. It is associated with rapid fluid resuscitation and rapid glucose correction. Risk is lower in adults.

SECTION B — Short Answer Questions

Answer ALL 5 questions. Write concise, structured answers.

- **SQ1.** List ALL 5 components of the VAP (Ventilator-Associated Pneumonia) Prevention Bundle.

- Elevation of the head of the bed to >30–45 degrees.
- Oral care (with chlorhexidine solution every 4–6 hours).
- Daily sedation vacation and assessment for weaning/extubation readiness.
- DVT (Deep Vein Thrombosis) prophylaxis — LMWH or pneumatic compression devices.
- GIT / Peptic ulcer (stress ulcer) prophylaxis — PPI or H2 blocker.

- **SQ2.** Describe the key steps in managing a patient with raised Intracranial Pressure (ICP).

- Elevate head of bed to 30–45 degrees.
- Keep head and neck in a straight, midline position — avoid jugular compression.
- Target Cerebral Perfusion Pressure (CPP) >70 mmHg. $CPP = MAP - ICP$. Maintain MAP and control ICP.
- Maintain normothermia — fever significantly increases cerebral metabolic demand and worsens ICP.
- Maintain normoglycemia — both hypoglycemia and hyperglycemia worsen neurological outcomes.
- Consider hyperosmolar therapy (Mannitol or Hypertonic Saline) for acute ICP spikes.
- Avoid hypercapnia — hyperventilation (target $PaCO_2$ 35–40 mmHg) can be used short-term to reduce ICP.

- **SQ3.** Differentiate between the FOUR types of shock with one example of each.

- **HYPOVOLEMIC** — Loss of circulating volume. Cause: hemorrhage (e.g., trauma, GI bleed), severe dehydration.
- **CARDIOGENIC** — Pump failure; the heart cannot maintain adequate output. Cause: massive MI, acute heart failure, cardiomyopathy.
- **OBSTRUCTIVE** — Mechanical blockage of blood flow. Cause: Tension Pneumothorax, Massive Pulmonary Embolism, Cardiac Tamponade.

- **DISTRIBUTIVE** — Massive vasodilation causing maldistribution of blood flow. Cause: Sepsis (most common), Anaphylaxis, Neurogenic shock.

- **SQ4.** Outline the management of Diabetic Ketoacidosis (DKA) including diagnosis, insulin dosing, and key complications.

- **DIAGNOSIS:** Random BG >250 mg/dL + pH <7.3 + Bicarbonate <15 mEq/L + ketonemia/ketonuria.
- **IV FLUIDS:** Start with 0.9% Normal Saline — 1 L over the first hour to restore circulating volume.
- **INSULIN:** Start continuous IV insulin infusion at 0.1 unit/kg/hr. Target BG reduction of 50–80 mg/dL/hr.
- **POTASSIUM:** Check serum K⁺ before starting insulin. Supplement if K⁺ <3.5 mEq/L. Insulin drives K⁺ intracellularly — monitor closely to prevent hypokalemia.
- **SWITCH TO DEXTROSE:** When BG drops to ~250 mg/dL, switch IV fluids to D5W + 0.45% NaCl to prevent hypoglycemia while continuing insulin to clear ketones.
- **KEY COMPLICATIONS:** Hypoglycemia, Hypokalemia (life-threatening arrhythmias), Cerebral Edema (leading cause of death especially in children — from rapid fluid/glucose correction).

- **SQ5.** Compare and contrast the blood pressure management in ISCHEMIC vs. HEMORRHAGIC stroke.

- **FIRST STEP FOR BOTH:** Non-contrast CT brain to differentiate hemorrhagic from ischemic stroke.
- **ISCHEMIC STROKE (first 48 hours):** DO NOT lower BP aggressively. Target BP ~180/105 mmHg. The elevated BP is compensatory, maintaining collateral flow to the ischemic penumbra. Exception: if IV thrombolytics (tPA) are given, lower BP to <185/110 before administration.
- **HEMORRHAGIC STROKE:** Lower BP more aggressively. Target systolic BP <140 mmHg to reduce hematoma expansion and re-bleeding risk.
- **MAJOR COMPLICATION of STROKE:** Aspiration Pneumonia — due to impaired gag reflex and dysphagia.
- **ECLAMPSIA (special case):** Treat BP >160/105 mmHg with IV labetalol or hydralazine. Magnesium sulfate for seizure prevention.